

Multiple Choice (20 marks)

1. There are 190 guests at a wedding. What is the least number of circular tables needed to seat all the guests if each table seats exactly 8 people?
 - (a) 22
 - (b) 23
 - (c) 24
 - (d) 25
2. If x and y are directly proportional and $x = 3$ when $y = 8$, what is the value of x when $y = 13$?
 - (a) 0.615
 - (b) 4.875
 - (c) 15
 - (d) 34.667
3. Two whole numbers have a least common multiple of 60. Each number is less than or equal to 12. The greatest common factor of the two numbers is 2. What are...
 - (a) 6 and 10
 - (b) 5 and 12
 - (c) 10 and 12
 - (d) 12 and 15
4. When $(x\sqrt{x^3})^4$ is simplified, what is the exponent of x ?
 - (a) 5
 - (b) 10
 - (c) 2
 - (d) 12
5. How many years are there in a decade?
 - (a) 5
 - (b) 10
 - (c) 15
 - (d) 20
6. What is the shortest distance between the y -axis and the point $(2, 7)$?

- (a) 1
 - (b) 2
 - (c) 3
 - (d) 7
7. What is the square of 15?
- (a) 15
 - (b) 15-Jan
 - (c) 225
 - (d) 256
8. Simplify 5^3 . Use a calculator, paper and pencil, or mental math.
- (a) 15
 - (b) 125
 - (c) 8
 - (d) 53
9. A circle passes through the points (3, 4) and (5, 7). Which of the following points CANNOT lie on the circle?
- (a) (-2, -1)
 - (b) (-1, -2)
 - (c) (5, 5)
 - (d) (6, 4)
10. Find the GCD of 25 and 55.
- (a) 5
 - (b) 11
 - (c) 50
 - (d) 275

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: An object moving along a line has velocity $v(t) = t \cos(t) - \ln(t + 2)$, where $0 \leq t \leq 10$. How many times does the object reverse direction?
2. True or False: Let $12 - 9 + c = 12$. What value of c makes the equation true?

3. True or False: Write 4.95 as a mixed number or fraction in simplest form.
4. True or False: Let $p, q,$ and r be constants. One solution to the equation $(x-p)(x-q) = (r-p)(r-q)$ is $x=r$. Find the other solution in terms of $p, q,$ and r .
5. True or False: For $T: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ where $T(1, 0) = 3$ and $T(0, 1) = -5$, find $T(-3, 2)$.
6. True or False: Jerry read a 200-page book in 10 hours. At that rate, how long will it take him to read a 320-page book?
7. True or False: Statement 1 — $4x - 2$ is irreducible over $\mathbb{Z}$. Statement 2 — $4x - 2$ is irreducible over $\mathbb{Q}$.
8. True or False: Find the order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15}) / \langle (1, 1) \rangle$
9. True or False: Find the product of the given polynomials in the given polynomial ring. $f(x) = 4x - 5, g(x) = 2x^2 - 4x + 2$ in $\mathbb{Z}_8[x]$.
10. True or False: Which of the following is most useful in establishing cause-and-effect relationships?

End of Paper